

Blosc frontiers: L40 and RTX 5070 Laptop

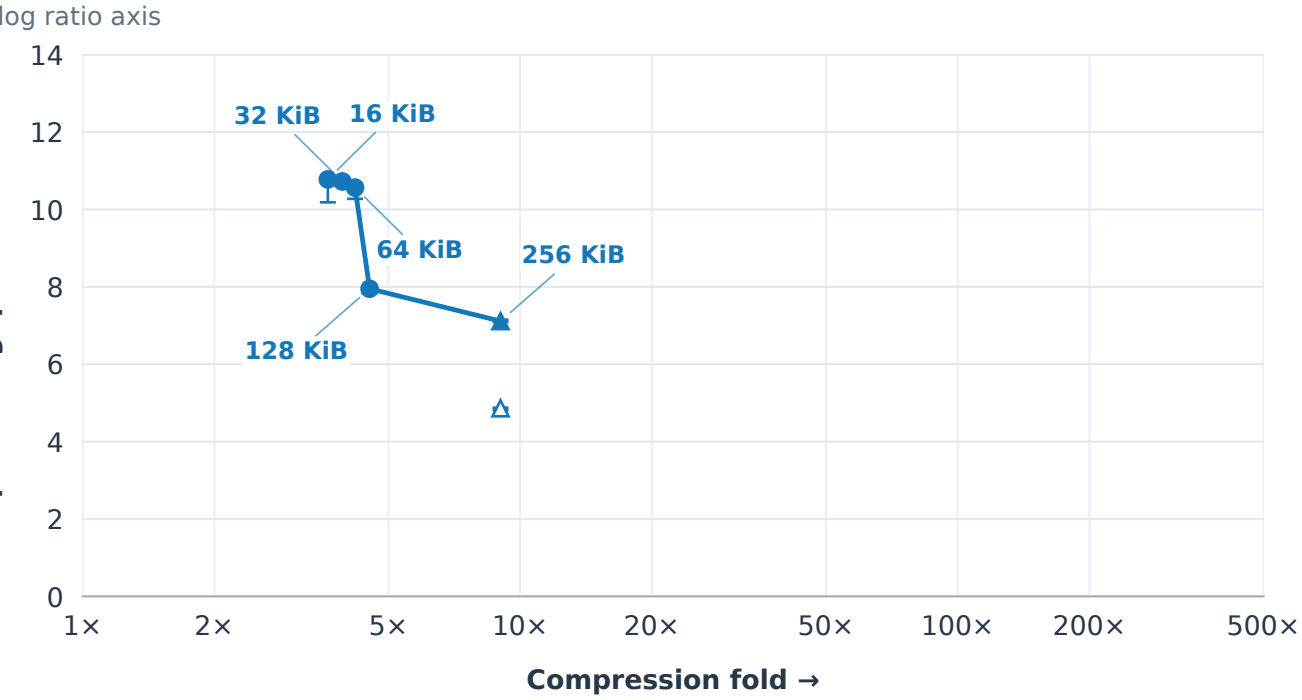
L40: 2026-09-06 · RTX 5070 Laptop: 2026-09-05 · nvCOMP 5.3

Different source revisions and toolchains: a comparison of measured setups.

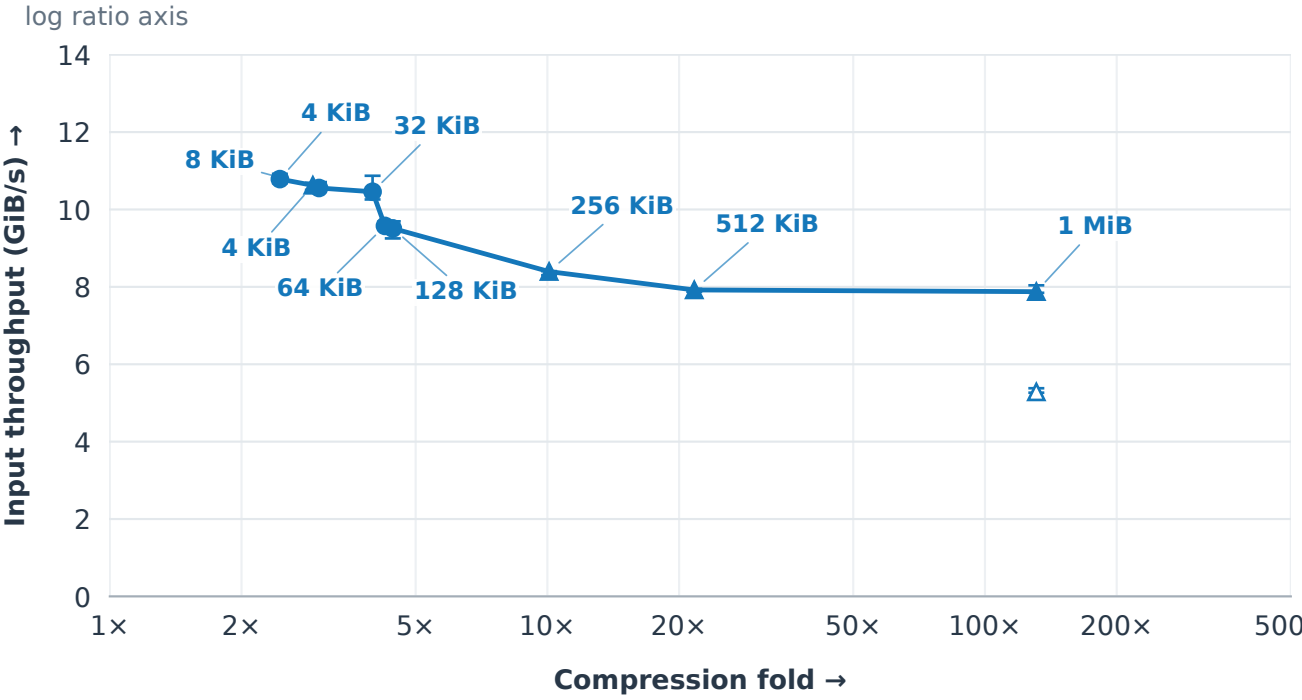
LZ4 Zstd no shuffle byte shuffle bitshuffle L40 RTX 5070 Laptop

Labels: L40 block size · bars: observed min-max · L40: 5 runs · RTX 5070 Laptop: 3 runs

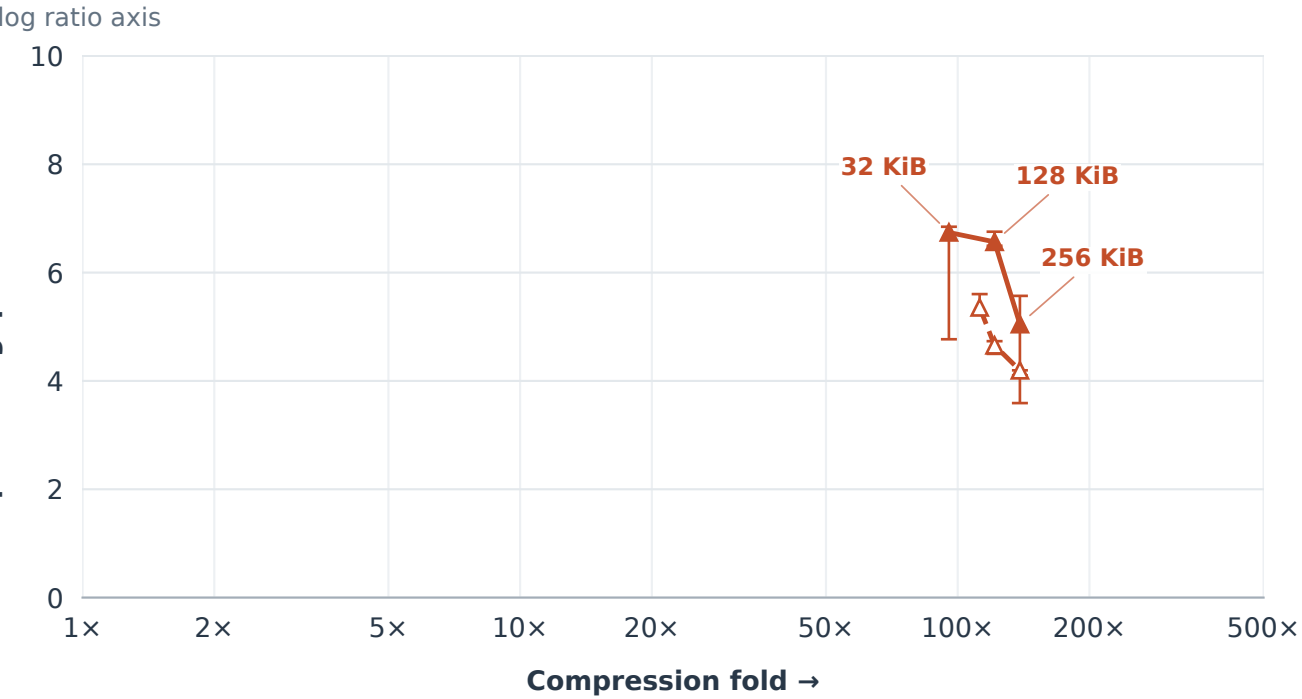
Repetitive XOR · 256 KiB Zarr chunks · LZ4



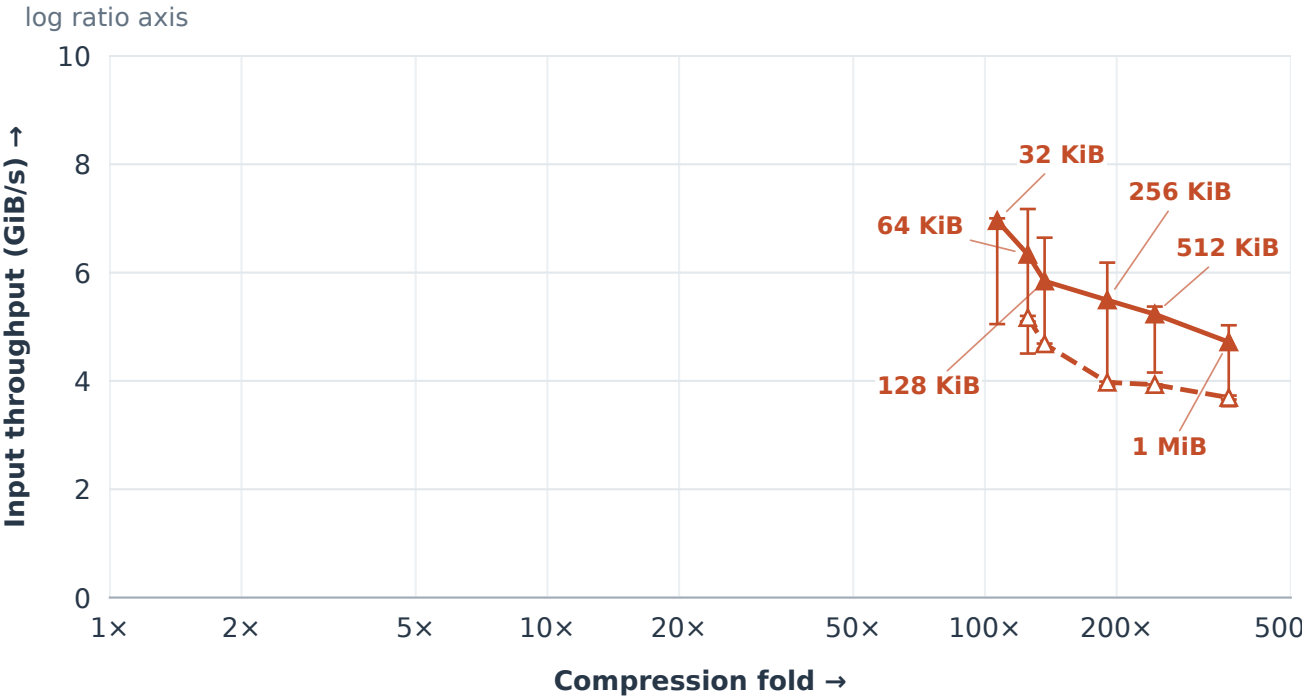
Repetitive XOR · 1 MiB Zarr chunks · LZ4



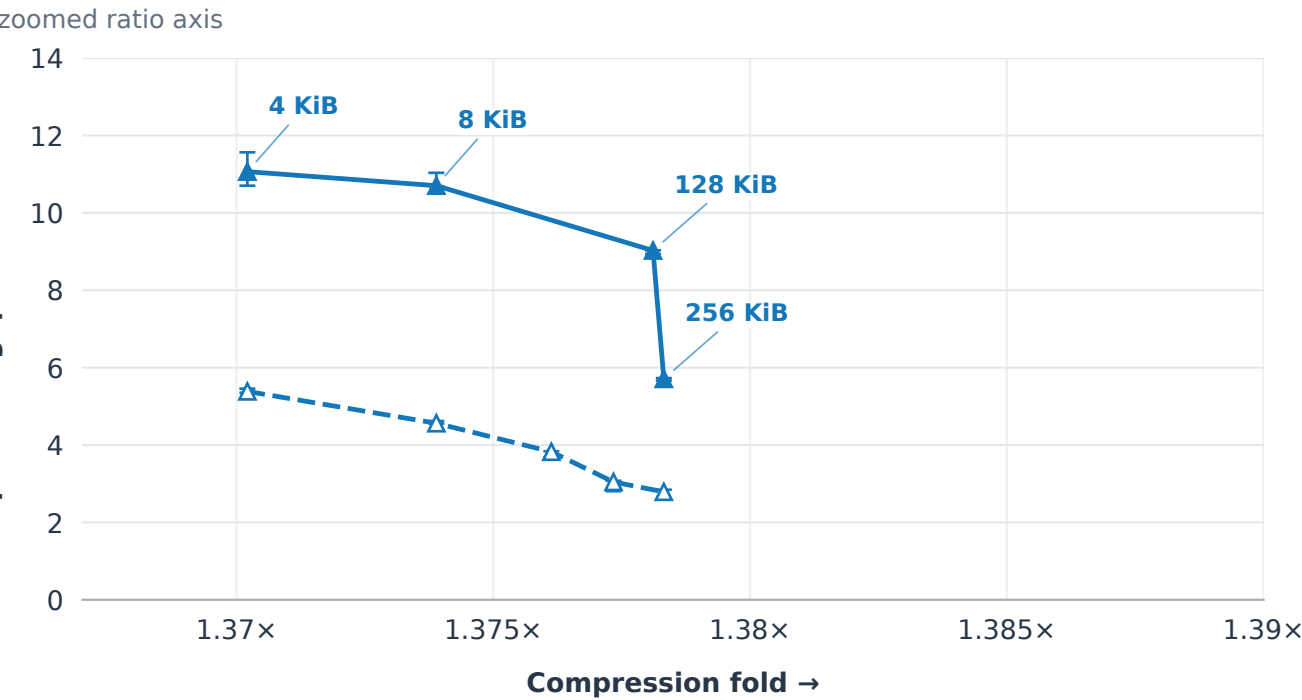
Repetitive XOR · 256 KiB Zarr chunks · Zstd



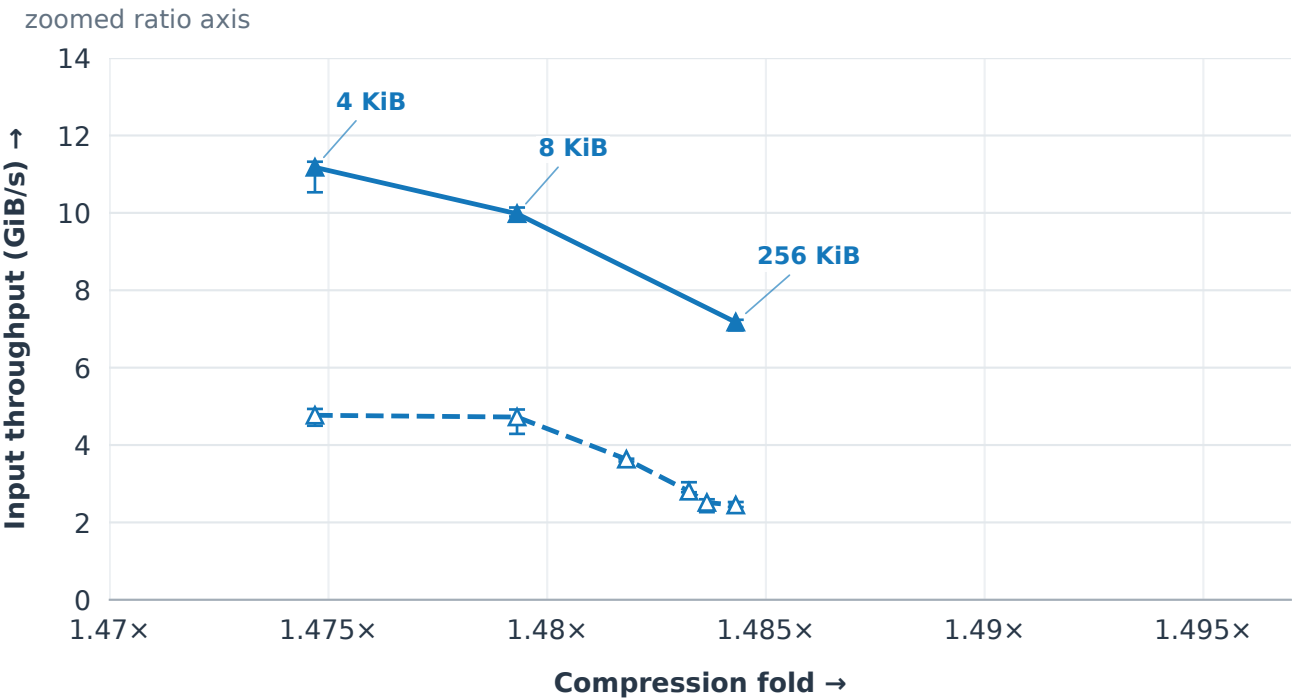
Repetitive XOR · 1 MiB Zarr chunks · Zstd



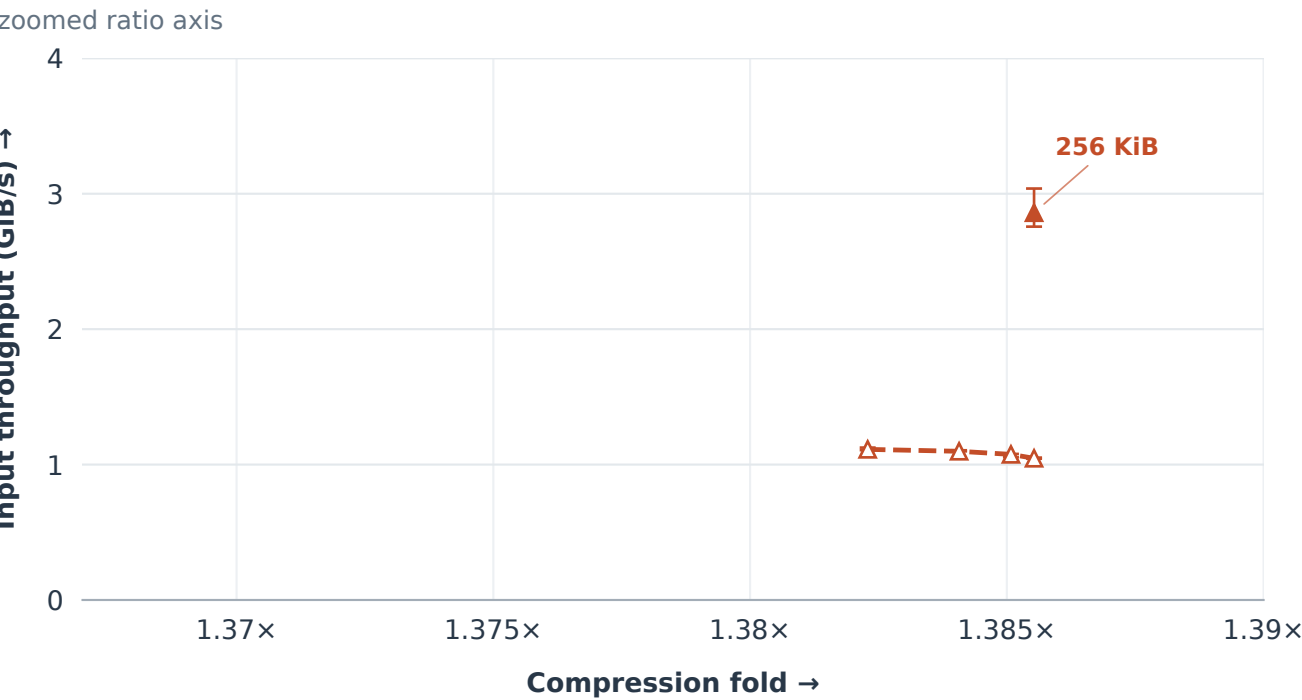
12-bit random · 256 KiB Zarr chunks · LZ4



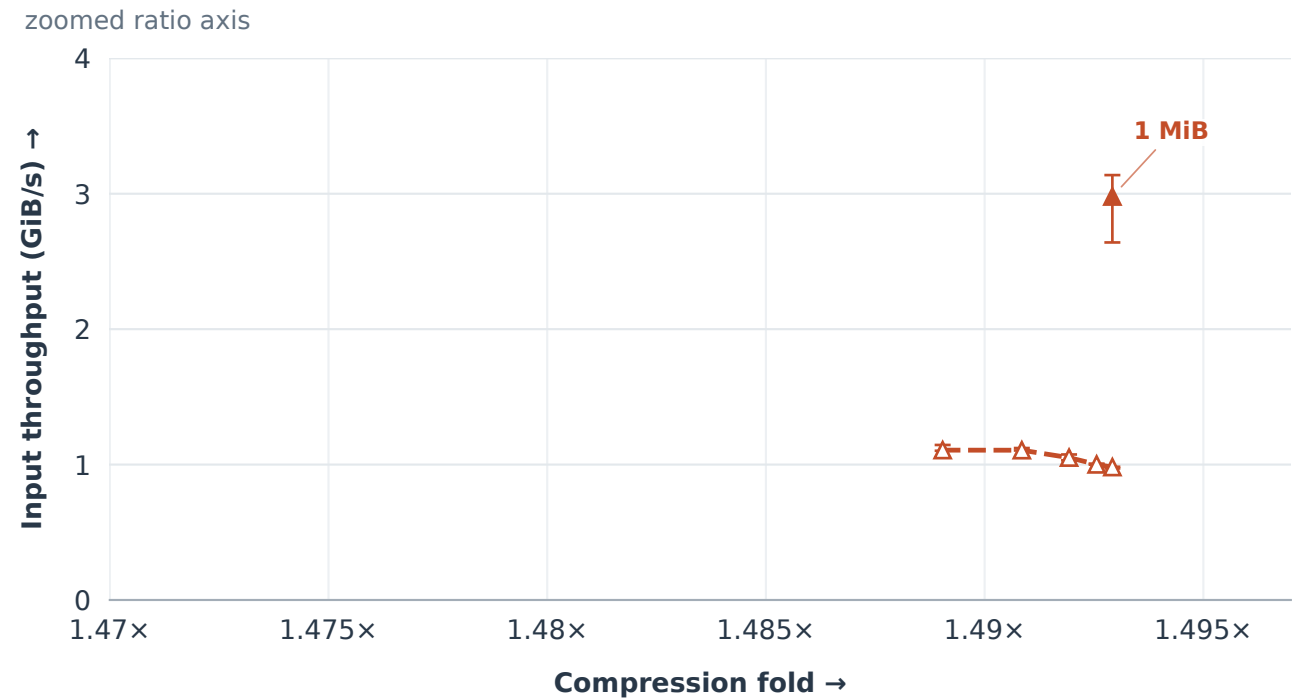
12-bit random · 1 MiB Zarr chunks · LZ4



12-bit random · 256 KiB Zarr chunks · Zstd



12-bit random · 1 MiB Zarr chunks · Zstd



Frontiers include all tested blocks and shuffles within each codec. Lines join measured settings.

Labels show L40 block sizes; the CSV and Markdown tables retain both sets of frontier identities.

Bars show observed spread, not confidence intervals. This comparison does not isolate the effect of the GPU.